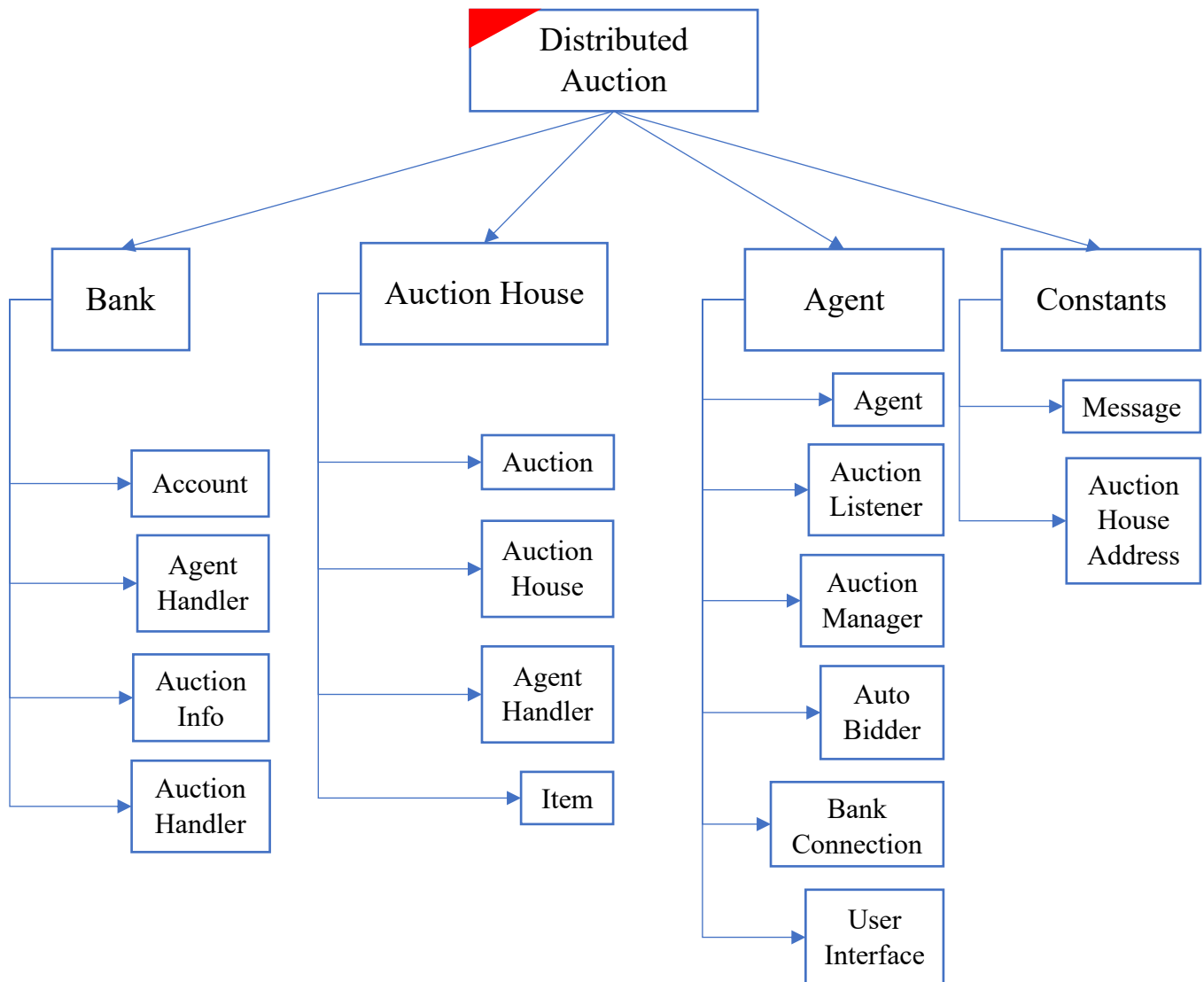


Proposed Design



Trigger: update from keyboard

Description of the Design

1. Bank Package

Bank Connection:

- Data:
 - socket: The socket connection with the Bank.
 - out: The ObjectOutputStream for sending messages to the Bank.
 - in: The ObjectInputStream for receiving messages from the Bank.
- Behavior:

- Establishes the connection with the Bank.
- Provides methods to create a new account and retrieve the list of available auctions from the Bank.
- Handles the communication with the Bank, including sending messages and receiving responses.

Agent Handler:

- Data:
 - socket: The socket connection with the agent.
 - account: The account object associated with the agent.
 - in: The ObjectInputStream for receiving messages from the agent.
 - out: The ObjectOutputStream for sending messages to the agent.
- Behavior:
 - Processes the messages received from the agent, such as creating a new account, sending the account balance, blocking and unblocking funds, and handling the termination of the agent.
 - Communicates with the Account and Bank classes to perform the requested operations.

Auction Handler:

- Data:
 - socket: The socket connection with the auction house.
 - account: The account object associated with the auction house.
 - in: The ObjectInputStream for receiving messages from the auction house.
 - out: The ObjectOutputStream for sending messages to the auction house.
- Behavior:
 - Processes the messages received from the auction house, similar to the AgentHandler.
 - Communicates with the Account and Bank classes to perform the requested operations.

Auction Info:

- Data:
 - accountNum: The account number associated with the auction.
 - auctionID: The unique identifier of the auction.
 - items: The list of items being auctioned.
- Behavior:
 - Stores the information about a specific auction, including the account number, auction ID, and the list of items being auctioned.

Bank:

- Data:
 - accountHashMap: A HashMap that stores the accounts of the agents and auction houses.
 - auctionHouseAddressHashMap: A HashMap that stores the addresses of the registered auction houses.
 - bankSocket: A ServerSocket that listens for incoming connections.
- Behavior:
 - Starts the bank server and listens for incoming connections from agents and auction houses.
 - Handles the creation of new agent and auction house accounts.
 - Manages the blocking and transfer of funds during the auction process.
 - Provides the list of active auctions to the agents.

2. Auction House Package

Agent Handler:

- Data:
 - socket: The socket connection with the agent.
 - auctionHouse: The reference to the auction house instance.
 - in: The ObjectInputStream for receiving messages from the agent.
 - out: The ObjectOutputStream for sending messages to the agent.
- Behavior:
 - Processes the messages received from the agent, such as requesting the list of active auctions, placing bids, and requesting the list of auctioned items.
 - Communicates with the AuctionHouse to perform the requested operations and sends the responses back to the agent.

Auction:

- Data:
 - auctionId: The unique identifier of the auction.
 - item: The item being auctioned.
 - currentBid: The current highest bid for the auction.
 - currentBidder: The account number of the current highest bidder.
 - active: A flag indicating whether the auction is still active.
- Behavior:
 - Manages the lifecycle of a single auction, including methods to place bids, retrieve the current bid and bidder, and close the auction.

AuctionHouse.java:

- Data:
 - activeAuctions: A collection of the active auctions hosted by this auction house.
 - items: A collection of the items being auctioned.
- Behavior:

- Hosts the auctions for the available items.
- Receives and processes bids from the agents.
- Communicates with the Bank to block and transfer funds during the auction process.
- Notifies the agents about the status of their bids (accepted, rejected, outbid).

Item.java:

- Data:
 - startingPrice: The starting price of the item.
 - currentBid: The current highest bid for the item.
 - currentBidder: The account number of the current highest bidder.
- Behavior:
 - Manages the information about a single item being auctioned, including the starting price, current bid, and current bidder.

3. Agent Package

Agent:

- Data:
 - accountNum: The unique account number of the agent.
 - bankConnection: The connection to the Bank.
 - auctionManager: The component responsible for managing the agent's interactions with the auction houses.
 - userInterface: The user interface for the agent.
 - autoBidder: The component responsible for automatically bidding on available auctions.
 - balance: The current balance of the agent's account.
- Behavior:
 - Connects to the Bank and creates a new account.
 - Retrieves the list of available auctions from the Bank.
 - Manages the connection and communication with the auction houses through the AuctionManager.
 - Allows the user to view the account balance, view available auctions, place bids, and set up automatic bidding.
 - Updates the account balance based on the auction outcomes.

Auction Listener:

- Data:
 - auctionSocket: The socket connection with the auction house.
 - agent: The reference to the agent instance.
 - in: The ObjectInputStream for receiving messages from the auction house.
 - running: A flag to indicate whether the listener is still running.
- Behavior:

- Continuously listens for and processes messages from the auction house, such as bid acceptance, bid rejection, auction win, and outbidding notifications.
- Updates the agent with the received auction-related information.

Auction Manager:

- Data:
 - agent: The reference to the agent instance.
 - auctionConnections: A HashMap that stores the socket connections to the auction houses.
 - outputStreams: A HashMap that stores the output streams for communicating with the auction houses.
 - listeners: A HashMap that stores the AuctionListener instances for each auction house connection.
- Behavior:
 - Establishes the connection to the auction houses and registers the agent with them.
 - Manages the AuctionListener instances for each auction house connection to receive updates.
 - Provides the ability to place bids on items through the auction houses.
 - Closes the connections to the auction houses when the agent is terminated.

AutoBidder:

- Data:
 - agent: The reference to the agent instance.
 - maxBids: A HashMap that stores the maximum bids for each auction.
 - currentBids: A HashMap that stores the current bids for each auction.
 - running: A flag to indicate whether the AutoBidder is still running.
- Behavior:
 - Periodically checks for available auctions and places bids on them, up to a maximum bid based on the agent's balance.
 - Updates the current bid information for each auction and checks if a new bid should be placed.
 - Stops the automatic bidding when the agent is terminated.

Bank Connection:

- Data:
 - socket: The socket connection with the Bank.
 - out: The ObjectOutputStream for sending messages to the Bank.
 - in: The ObjectInputStream for receiving messages from the Bank.
- Behavior:
 - Establishes the connection with the Bank.

- Provides methods to create a new account and retrieve the list of available auctions from the Bank.
- Handles the communication with the Bank, including sending messages and receiving responses.

4. Constants Package

AuctionHouseAddress.java:

- Data:
 - ipAddress: The IP address of the auction house.
 - portNum: The port number of the auction house.
- Behavior:
 - Provides methods to get the IP address and port number of the auction house.
 - This class is used to store and share the address information of the auction houses with the agents.

Message.java:

- Data:
 - args: An ArrayList of serializable objects, where the first element represents the command, and the rest are the arguments.
- Behavior:
 - Provides a constructor to create a new Message object with the specified command and arguments.
 - Includes methods to get the command, convert the message to a string, and extract specific arguments from the message.
 - This class is used as the communication medium between the various components (Bank, AuctionHouse, and Agent) in the system.